

Intake:	2020 onwards		Specialization:	Civil Engineering			
Semester	Code	Module Title			Credits	C/E/O	GPA / NGPA
1	CE1023	Fluid Mechanics			2.0	C	GPA
Hours/Week		Prerequisites / Corequisites			Evaluation %		
Lecture	Lab/Tutes				CA		WE
2	2/4	None			20	80	
Learning Outcomes							
After completing this module, students should be able to:							
1. describe fluid properties and characteristics of fluids in reference to engineering practice and related applications, 2. demonstrate hydrostatic forces on structures such as dams and radial gates and analyse the stability of floating vessels, and 3. apply concepts of conservation of mass, energy and momentum of fluids in applications such as determining discharge and energy losses in pipelines and forces on bends/vanes.							
Syllabus Outline							
1. Introduction 2. Fluid properties 3. Hydrostatic pressure 4. Buoyancy 5. Relative equilibrium 6. Fluids in motion 7. Hydraulic machinery							

Intake:	2020 onwards	Specialization:	Computer Science & Engineering		
Semester	Code	Module Title	Credits	C/E/O	GPA / NGPA
1	CS1033	Programming Fundamentals	3.0	C	GPA
Hours/Week		Prerequisites / Corequisites	Evaluation %		
Lecture	Lab		CA	WE	
2	2	None	20	80	
Learning Outcomes					
After completing this module, students should be able to					
•Device algorithms to solve simple computational problems •Develop programs from algorithms using a high-level programming language (e.g., Python) •Develop programs for simple control applications using embedded hardware platforms					
Syllabus Outline					
•Introduction to Computing •Python: Introduction, Operators, Expressions •Python: Selection Control Structures •Python: Loop Control Structures SP •Python: Lists •Python: Functions •Data Representation •Problem Solving I •Problem Solving II •Problem Solving III •Computer System & Hardware I •Computer System & Hardware II					

Intake:	2020 onwards	Specilization:			
Semester	Code	Module Title	Credits	C/E/O	GPA / NGPA
1	EE1040	Electrical Fundamentals	2.0	C	GPA

Hours/Week		Prerequisites / Corequisites	Evaluation %	
Lecture	Tute/Lab		CA	WE
2	2/4	None	20	80

Learning Outcomes

After completing this module, the student should be able to;

- Describe the practical aspects of basic circuit elements.
- Analyze ac circuits using series/parallel simplifications, voltage/current division rules.
- Solve three-phase balanced circuits in terms of line quantities and power.
- Draw up a complete wiring circuit of a household and appreciate the importance of different protection and safety devices

Syllabus Outline

1. Overview of Electrical Engineering

2. Basic Circuit Elements

Physical characteristics of linear circuit elements (resistors, inductors and capacitors), voltage-current relationships, voltage sources, solutions of resistive circuits using Kirchoff's laws.

3. AC Theory

Sinusoidal waveform, waveform parameters, phasor representation, complex representation, impedance, admittance, complex power and energy, power factor, series/parallel simplifications, voltage/current division rules, AC circuit calculations.

4. Three Phase Balanced Circuits

Definition of balanced three phase systems, circuit diagrams, delta-star connection and transformation, per-phase equivalent circuit, power factor correction.

5. Circuit Protection and Basic Electrical Safety

Basic components of a domestic electrical system, overcurrent/short circuit protection, earth leakage protection, devices, case studies

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
1	MA1014	Mathematics	3	C	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
5/2	1/1	None	20	80	

Learning Outcomes

After the successful completion of this course, students should be able to

- Identify basic operations and functions of complex variables, explore 3D geometry using vectors and solve basic eigenvalue problems for matrices.
- Use real functions of one real variable up to power series.
- Solve Differential Equations up to second order linear with non-constant coefficients.

Syllabus Outline

Algebra

- **Complex Numbers:** Euler's Identity, complex valued functions and branches.
- **Vectors:** vector algebra, vector product, scalar product, scalar triple product, vector triple product, equations of line and plane, vector norms
- **Matrices:** transpose, adjoint, determinant, inverse and trace of a matrix, system of equations, Cramer's rule, Gaussian elimination, echelon forms, rank, eigen values and eigen vectors, diagonalization, matrix norms.

Real Analysis

- **Sets and Inequalities:** Introduction to quantifiers and sets, real number system, inequalities, supremum and infimum, completeness axioms.
- **Functions, Limits and Differentiability:** relations, functions and their inverses, limit of a function, continuity, differentiability
- **Basic Theorems:** Intermediate value theorem, extremum value theorem, Rolle's theorem, mean value theorem, L' Hopital's rule.
- **Sequences and Series:** Convergence of sequences and series, monotone convergence theorem. Power series, Taylor's series.

Integration and ODE

- **Riemann integration:** integral as an area, First and second fundamental theorems of calculus, Leibniz rule, Integrability of a continuous function, Integration by parts, mean value theorem for integrals, Improper integrals: tests of convergence, gamma function.
- **Ordinary differential equations:** classification of ODEs (Linear and non-linear), First order ordinary differential equations: variable separable, homogeneous, linear, Bernoulli Second order linear differential equations: equations with constant coefficients, Wronskian method

Intake:	2020 onwards	Specialization:			
Semester	Code	Module Title	Credits	C/E/O	GPA / NGPA
1	ME1033	Mechanics	2.0	C	GPA
Hours/Week		Prerequisites / Corequisites	Evaluation %		
Lecture	Tute/Lab		CA	WE	
2	2/4	None	20	80	

Learning Outcomes

Upon completing this course, the students should be able to:

- Calculate sectional properties of plane areas,
- Calculate internal forces in beams,
- Identify statically determinate / indeterminate trusses, their stability and determine forces in truss members.

After completing this part (Dynamics) of the module, the students should be able to:

- Analyse the geometry of motion of particles, rigid bodies and 2D linkages,
- Determine forces and energy associated in particles and rigid bodies in motion,
- Analyse natural vibrations of damped, single degree of freedom systems.

Syllabus Outline

Statics

- Properties of plane areas
- Internal forces (BMD & SFD)
- Principle of superposition
- Determination of forces in assemblies of rigid bodies

Dynamics

Fundamentals of Dynamics

- Kinematics of particles (rectilinear and curvilinear motion, relative motion, general motion in 2D) and rigid bodies (relative motion between two points in a rigid body, velocities in 2D link mechanisms, instantaneous centre of rotation method, introduction to acceleration)
- Kinetics of particles and rigid bodies (force, torque, work, energy and power, linear momentum, angular momentum)

Mechanical Vibrations

- Free vibrations (undamped and damped) of single degree of freedom systems.

Intake:	2020 onwards		Specialization:			
Semester	Code	Module Title		Credits	C/E/O	GPA / NGPA
1	MT1023	Properties of Materials		2.0	C	GPA

Hours/Week		Prerequisites / Corequisites	Evaluation %	
Lecture	Tute/Lab		CA	WE
2	2/4	None	20	80

Learning Outcomes

At the completion of this module, students should be able to ;

- Recognize the structure of metals, polymers and ceramics
- Identify the relationships between the structure of materials, their properties and applications
- Assess the properties of engineering materials

Syllabus Outline

- Introduction to engineering materials
- Structure of atoms, atomic theories, atomic bonding in materials
- Crystal structures and defects
- Introduction to nanomaterials
- Mechanical properties of materials
- Electrical properties of materials
- Degradation of Materials
- Functional Materials and their applications
- Basic materials selection

Intake	2020 onwards	Specialisation	Computer Science & Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
2	CS2813	Visual Programming		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
1	2	2.0	None	60	40
Learning Outcomes					
After completing this module, students should be able to 1. Develop a working program for specified programming problems using a visual programming environment.					
Syllabus Outline					
Outline Syllabus 1. Introduction to the concept of visual programming. 2. Introduction to visual programming environments. 3. Practice of visual programming using .NET Framework Objects, Properties, Events and Methods; Variables, Data Types and Controls; Use of Forms and Controls to create User Interfaces; Program Control Flow; String and file manipulation; Arrays; Procedures and Functions; Exception Handling; Database Programming					

Intake:	2020 onwards	Department:	Languages		
Semester	Code	Module Title	Credits	C/E/O	GPA / NGPA
2	EL1030	Language Skills Enhancement	2.0	C	GPA
Hours/Week		Prerequisites / Corequisites	Evaluation %		
Lecture	Tute/Lab		CA	WE	
	2	None	100	0	

Learning Outcomes

At the completion of this module, students at elementary, intermediate and advanced proficiency at entry in terms of language skills relevant for engineering undergraduates, will be able to:

- 1 demonstrate having achieved the competencies for listening defined by UTEL bands 6, 7 and 8 respectively.
- 2 demonstrate having achieved the competencies for speaking defined by UTEL bands 6, 7 and 8 respectively.
- 3 demonstrate having achieved the competencies for reading defined by UTEL bands 6, 7 and 8 respectively.
- 4 demonstrate having achieved the competencies for writing defined by UTEL bands 6, 7 and 8 respectively.

Syllabus Outline

- 1 Speaking on given topics.
- 2 Asking questions and responding to questions.
- 3 Reading comprehension
- 4 Summarising and synthesising
- 5 Describing objects, mechanisms and processes
- 6 Discussion/ writing activities
- 7 Describing data and graphical information
- 8 Functional grammar

Intake	2020 onwards	Specialisation			
Semester	Code	Module Title		C/E/O	GPA / NGPA
2	EN1803	Basic Electronics for Engineering Applications		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
2	2	3	None	40	60
Learning Outcomes					
LO1 Design simple electronic circuit s using diodes and transistors. LO2 Design simple combinational logic circuits. LO3 Use operational amplifiers in electronic instrumentation applications. LO4 Design simple electronic instrumentation systems using sensors, operational amplifier circuits, data converters, and microcontrollers. LO5 Identify noise and interference sources in electronic instrumentation systems.					
Syllabus Outline					
1. Introduction to Basic Electronic Devices Historical aspects, electronic systems, modern electronic industry, electronic manufacturing process, software tools.					
2. Microcontrollers Introduction, applications, internal peripherals, programming, communication, communication with a computer, connecting external modules.					
3. Sensors, Transducers, and Actuators Performance characteristics of transducers: dynamic range, sensitivity, resolution, input/output impedance, useful frequency range, time constant and steady state error, sensor/transducer types and examples, actuators, component selection.					
4. Operational Amplifiers Need of op-amps in interfacing sensors, operation and characteristics, non-inverting and inverting configuration. Applications: inverter, comparator, voltage follower (buffer), adder, subtractor, integrator, differentiator, oscillator, component selection.					
5. Electronic Instrumentation Systems Analog-to-digital and digital-to-analog conversion, frequency ranges and bandwidth, signal reflection in cables, noise and interference, noise reduction methods.					
6. Diodes, Diode Circuits, and Applications Operation and characteristics of a p-n junction and diode. Zener diode, light emitting diode, and other types (varactor, Schottky, and PIN). Rectification, clamping and limiting circuits, a simple power supply design example.					
7. Bipolar Junction Transistors (BJTs), Metal-Oxide Semiconductor Field Effect Transistors (MOSFETs), and Circuits Operation and characteristics of the BJT/MOSFET, using the transistor as a switch and an amplifier, biasing schemes, amplifier configurations and parameters, selected integrated circuits (e.g., 78xx, 79xx, 555, 2577, 2596), a power supply design example, protection.					
8. Logic Gates and Circuits Logic gates, review of Boolean algebra, minimization of logic expressions using Karnaugh maps, combinational logic circuits (half adder, full adder, comparator, encoder, decoder, multiplexer and demultiplexer).					

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
2	MA1024	Methods of Mathematics	3	C	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
5/2	1/1	MA1014	30	70	
Learning Outcomes					
After the successful completion of this course students should be able to • Understand the basic numerical techniques and its applications • Understand the basic theories on severable variable calculus and its applications • Understand the basic concepts of Probability and Statistics and identify the applications of them					
Syllabus Outline					
Numerical Methods • Numerical solution of non-linear equations: Bisection, Fixed point, Newton’s methods. • Numerical differentiation and integration: Trapezoidal and Simpson’s rules. • Interpolation: Lagrange polynomial, Newton’s Divided Difference and Least Square method. • Numerical solution to systems of equations: Gauss-Jacobi Gauss-Seidel methods • Numerical solution to ODEs-Euler’ s method • Numerical Optimization-Golden section search, method of finding roots of the derivative					
Multivariate Calculus and Introductions to PDEs • Limits, Continuity, Partial Derivatives, Mean Value Theorem, Differentiability • Chain rule-Bivariate • Gradient, Tangent plain, Directional derivatives • Jacobian, Hessian • Inverse Function Theorem, Implicit Function Theorem. • Maxima, minima and Saddle points, Lagrange multipliers • Taylor series expansion for two variable, quadratic forms • Double integrals: Fubini’s theorem, Change of variables, polar coordinates • Solution of the exact ODE • Introduction and solve first order PDE, Solution by the method of Characteristics					
Probability and Statistics • Random experiments and random events • Concept of probability, computing probability using two-way tables • Conditional probability and Bayes theorem. • Discrete and continuous random variables and their properties • Distributions and their applications (Bernoulli, Binomial, Poisson, Uniform and Normal distribution) • Normal approximation for Binomial & Poisson and their applications • Student’s t–distribution • Estimation of parameters: point estimators and their properties • Central Limit Theorem and its applications • Confidence intervals for mean • Covariance and correlation coefficient					

Intake:**2020 onwards****Specialization:****Mechanical Engineering**

Semester	Code	Module Title	C/E/O	GPA / NGPA
2	ME1091	Engineering Drawing & Computer Aided Modelling	C/E	GPA

Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tute			CA	WE
1	4	3	None	100	0

Learning Outcomes

After completing this module, the students should be able to:

LO-1: Demonstrate knowledge of projection methods

LO-2: Apply principles of orthographic projections in drawing views of a given component

LO-3: Make use of graphical construction techniques in engineering drawings

LO-4: Utilize CAD software to draw orthographic projections, printing, and plotting

LO-5: Develop 3-dimensional models in a virtual environment

Syllabus Outline**Section 1: Projection Methods**

Orthographic and isometric projections.

Section 2: Orthographic Views

Orthographic views of objects from a given isometric view; Third view from two orthographic views; Sectional views; Orthographic views of an assembled object; Orthographic views of an assembly of a set of given components.

Section 3: Graphical Constructions

Lines and planes: graphical estimation of true lengths, inclinations, traces, auxiliary projection methods & true shapes of sections; Interpenetration curves: construction of interpenetration curves of cylinder, cone, sphere, pyramid; Developments: construction of developments of prism, cylinder, cone, pyramid, and developments by the method of triangulation.

Section 4: Computer aided drafting and modelling

Software tools and techniques for drafting and modelling.

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
2	MT1070	Thermodynamics and Phase Equilibria		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. State the laws of thermodynamics. 2. Express equilibrium and equilibrium conditions. 3. Describe state functions and process variables 4. Apply the basic integration and differentiation rules to thermodynamic problems. 5. Use general strategy for deriving thermodynamic relations. 6. Calculate equilibrium conditions between solid-liquid, solid -gas, Liquid-gas phases via the Clapeyron equation					
Syllabus Outline					
• Laws of classical thermodynamics Thermodynamic laws, concept of work, heat enthalpy, entropy, and their relationships, Clausius inequality, Concept of free energy, thermodynamic variables, thermodynamic relationships, Coefficient & Maxwell relations. • Phase Equilibria Heat and Work Exchanges in Solid-Liquid Transformations, Gibbs phase rule, Clausius–Clapeyron equation, H-T, diagrams, Thermal Supercooling and Metastability, Reversible & Irreversible Solidification. • Statistical Thermodynamics Basic Concepts, Microstate Macrostate & entropy, Statistical thermodynamic conditions for equilibrium, Constrained maximum in the entropy function, evaluation of macroscopic properties from partition function. • Thermodynamics of Alloys Entropy of mixing, ideal and regular binary solution, Derivation of binary phase diagrams from solution model, Free energies of binary and ternary phase diagrams. • Phase diagrams in metallurgy Binary phase diagrams, Eutectic, peritectic and other systems, Formation of microstructure in metals, Interpreting phase diagrams, Lever rule, Introduction to iron-carbon phase diagram. • Thermodynamic effects of external electric fields Equilibrium within an electrolyte, Equilibrium in two phase-systems involving an electrolyte, Equilibrium in an electrochemical cell, General Galvanic cell, Temperature dependence of E.M.F.					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
2	MT1080	Fundamentals of Materials Science and Engineering		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
4.0	2/6	4.0	None	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Recognize the duties and responsibilities of a materials engineer within industrial and social contexts. 2. Relate the properties of main classes of engineering materials to the type of bonding and defects present. 3. Explain the structure of materials through fundamental crystallographic concepts. 4. Discuss structure, properties and applications of non-crystalline, semicrystalline and liquid crystalline materials. 5. Describe functional properties of materials and explain their applications in advanced functional devices. 6. Recognize the use of appropriate standards and specifications for materials testing and design.					
Syllabus Outline					
• Introduction Historical perspectives of materials science and engineering; The materials paradigm; Role of materials in manufacturing industry; Role of the materials engineer in the industry • Bonding forces and energies Bonding forces and energies of primary and secondary bonding; Correlation of the interatomic bonding to materials properties of main classes of engineering materials • Crystalline state Crystals and lattices: Definitions of a crystal and lattice points, fractional coordinates, unit cell calculations; Symmetry; Crystal systems and geometry; Reciprocal lattice, Ewald sphere, crystal vibrations, Brillouin zone • Defects in crystals • Semicrystalline, non-crystalline (amorphous) and liquid crystalline state Macromolecular structure; Amorphous materials; Liquid crystalline state • Introduction to functional materials and devices Optical/ Magnetic/Electrical/Electronic properties of materials; Bio-materials, Energy materials, Materials for sensors and actuators • Concept of standards and specifications • Economic, environmental and societal issues related to materials science and engineering					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
2	MT1940	Fundamentals of Engineering Design and Workshop Practice		C	NGPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
1.0	4.0	3.0	none	100	-
Learning Outcomes					
At the completion of this module, students should be able to; 1. Identify and define a design problem 2. Analyse product development strategies as applicable to engineered commodities 3. Create and clearly communicate the conceptual design 4. Organize strategies for effective marketing of designed products 5. Use common engineering materials, tools, equipment, and processes 6. Work collaboratively with others and articulate practices that lead to successful teamwork					
Syllabus Outline					
• Engineering design process- Steps in engineering design • Problem definition and need Identification-Stakeholder analysis • Conceptual design process • Introduction to creative thinking- Models of the brain and creativity, creative thinking methods • Decision making and concept selection methods- Decision tree, weighted decision matrix • Prototype specifications • Societal considerations in engineering design- Safety in design • Case Studies • Workshop Practice • Report writing					

Semester	Code	Module Title		C/E/O	GPA/NGPA
3	EE2804	Applied Electricity		C	GPA
Hours/Week		Credits	Pre-Requisites / Co-Requisites	Evaluation (%)	
Lecture	Lab/Tutorial			CA	WE
2	2	3	None	40	60

Learning Outcomes

After completing this module, the student should be able to;

1. Assess the performance of transformers and different electric motors under a variety of load conditions.
2. Select a suitable electric motor for a given application.
3. Describe the application of electrical power in lighting, heating and welding.
4. Apply wiring regulations for domestic wiring and cable selection .
5. Apply cost minimization strategies in the utilization of electric power.

Syllabus Outline

1. Transformers

Single Phase transformers, EMF equation, equivalent circuit & phasor diagram, losses & efficiency, voltage regulation, test on transformers, use of three phase transformers.

2. Induction Motors

Types of rotors and windings, induction motor action, torque speed characteristics, losses and efficiency, starting and speed control, ratings and applications, single phase induction motors and their applications.

3. DC Machines

Equivalent circuits, motor and generator operation, characteristics of series, shunt and compound motors, starting and speed control, industrial applications, brushless DC motors

4. Special Purpose Motors

Universal Motors: constructional and operational characteristics.
Stepper Motors: operation and types, applications.

5. Solid State Control

Introduction to solid state control of dc and ac motors, principles of four-quadrant operation.

6. Electric Lighting

Basic principles, characteristics of light, lamps and luminaires, average lumen method of lighting calculations.

7. Heating and Welding

Heating: heating methods (Joule, induction and dielectric heating), industrial applications.
Electric Welding: types, requirements, welding transformers.

8. Electrical Wiring

Wiring regulations, circuits and wiring symbols, selection and voltage drop calculations of cables, earthing.

9. Economics of Power Utilisation

Cost of Electric Power: fixed, variable and maximum demand charges, tariffs.
Demand Management: power factor correction.

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
3	MA2014	Differential Equations	2	E	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
2	0	MA1024	30	70	

Learning Outcomes

After the successful completion of this course students should be able to

- Demonstrate the knowledge of Laplace Transform and its applications
- Apply Fourier series to transform functions to trigonometric infinite series
- Classify Partial Differential Equations (PDE) and reduce PDEs to canonical form

Syllabus Outline

Ordinary Linear Differential Equations with Variable Coefficients

- Solutions in series form, Frobenius method.
- Special functions: Introduction of Legendre Polynomials and Bessel's functions.

Laplace Transform and Application to DE

- Laplace transforms of elementary functions and some basic theorems on Laplace transform.
- Inverse Laplace transform, methods to find inverse transform, Convolution theorem.
- Application of Laplace transforms to find solutions to ODEs and systems of ODEs.
- Transfer functions, concepts of stability and controllability.
- Complex Inversion formula.

Fourier Series

- Fourier coefficients, Dirichlet's condition, odd and even functions. Half range series.
- Parseval's Theorem
- Complex Fourier Series
- Fourier series as the norm minimizer

Partial Differential Equations

- Canonical Forms
- Classification of second-order partial differential equations: Hyperbolic, Parabolic and Elliptic
- Linear and Nonlinear First Order Partial Differential Equations. Equations solvable by direct integration
- Solutions by separation of variables.
- Fourier series application to boundary value problems.
- Solve partial differential equations by using Laplace transform.

Fourier Transform and Application to PDE

- Fourier integral theorem
- Fourier transforms, Fourier sine and cosine transform
- Complex form of Fourier Integral
- Properties of Fourier transform.
- Application of Fourier Transform to solve PDEs

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
3	MA2024	Calculus	2	E	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
2	0	MA1024	30	70	
Learning Outcomes					
After the successful completion of this course students should be able to • Perform vector differentiation and integration and evaluate vector and scalar quantities • Perform contour integration techniques to find the complex integrals • Apply conformal mapping					
Syllabus Outline					
Vector Calculus • Vector Functions. • Vector differentiation and differential operators. • Space curves and curvature, line integrals • Greens’ Theorem • Surfaces and curvature, surface integrals • Curvilinear coordinate systems: spherical and cylindrical coordinates • Stokes’ theorem. • Triple integrals. • Divergence theorem					
Complex Variables • Complex valued functions and branch cuts. • Analytical function and Cauchy-Reimann equation. • Complex Integration • Cauchy’ s integral formula. • Singularities, zeros and poles. • Taylor and Laurent series. • Residue theorem and applications of residue theorem for real integrals • Introduction to Conformal mapping.					

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
3	MA3024	Numerical Methods	2	E	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
2	0	MA1024	30	70	
Learning Outcomes					
After the successful completion of this course students should be able to • Demonstrate knowledge and understanding of common numerical methods • Apply numerical methods to obtain approximate solutions • Compare different numerical methods using efficiency and accuracy of the solution					
Syllabus Outline					
Solving Systems of Linear Equations • Condition of a linear System • Direct methods • LU decomposition • Symmetric, Positive definite Matrix decomposition • Tri diagonal Matrix decomposition Numerical Solutions of System of Non-Linear Equations • Fixed Point iteration • Newton’s Method Curve Fitting • Splines Numerical Optimization • Multi-dimensional • Steepest decent Numerical Solutions of Initial Value Problems for of ODEs ▪ Single step methods ▪ Euler’s Method & Modified Euler Method ▪ Runge-Kutta methods • Numerical Solutions of a System of IVP s • Multi-Step methods ▪ Adams methods (implicit, explicit) ▪ predictor-corrector methods Numerical solution of Linear & Nonlinear BVP s • Shooting Method • Finite Difference Approximation					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
3	MT2021	Polymer Science and Technology		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Name, classify and describe specific features of polymers. 2. Identify & illustrate isomeric forms of polymers. 3. Discuss amorphous and crystalline arrangement of polymer chains. 4. Calculate molecular weights of a given polymer. 5. Describe principal steps, mechanisms, and special features of different types of polymerisations, copolymerisations. 6. Discuss industrial polymerisation techniques. 7. Describe thermal transitions observable in polymers. 8. Identify a suitable polymeric material for a given application. 9. Describe the function of additives used in rubber compounding. 10. Develop polymer compounds to satisfy product service Requirements. 11. Discuss processability of rubber compounds. 12. Describe polymer product manufacturing technologies and choose a suitable manufacturing technique to turnout a given polymer product. 13. Identify design features of polymer products.					
Syllabus Outline					
• Basic concepts, Classification & Nomenclature - Naming, classification and description of specific features of polymers • Isomerism in polymers - Identification & illustration of various isomeric forms of polymers • Polymer morphology - Discussion of amorphous and crystalline arrangement of polymer chains and description of factors contributing for crystallisation of polymers • Polymer molecular weight - Calculation of different molecular weight averages of a given polymer • Polymerisation reactions - Explanation of chemistry and specific features of different types of polymerisation reactions including the consequences of chain transfer, auto-acceleration inhibition and retardation Application of Carother's equation in characterising step reaction polymerisations Description of principal steps and mechanisms involved in different types of chain reaction polymerisations • Copolymerisations - Discussion of composition drift observable in copolymerisation reactions and reactivity ratios of co-monomers • Industrial polymerisation techniques - Description of different polymerization techniques • Thermal transitions - Explanation of thermally assisted phase transformations in polymers • Structure property relationships and applications - Description of General purpose polymers and Special purpose polymers • Additives used in polymer compounding - Discussion on ingredients used in polymer compounding • Rubber compounding processes - Description on designing polymer compounds to satisfy					

product service requirements and Techniques used for preparing rubber compounds

- **Testing of polymer compounds** - viscometry, Rheometer curves etc.
- **Shaping operations in rubber technology and Plastics technology** - Description of polymer product manufacturing technologies and choose a suitable manufacturing technique to turnout a given polymer product.
 - Extrusion
 - Injection Moulding
 - Compression Moulding
 - Transfer Moulding
 - Blow Moulding
 - Thermoforming
 - Film Blowing
 - Rotational Moulding
 - Calendaring

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
3	MT2053	Communication Skills		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
1.0	2.0	2.0	None	100	0
Learning Outcomes					
At the completion of this module, students should be able to; 1. Demonstrate the ability of speaking with confidence in the presence of superiors and peers in an organized manner. 2. Enhance the skill of Listening and comprehending effectively in lectures / presentations 3. Develop reports that present and defend a clear, precise thesis using effective evidence, a variety of sources, and appropriate documentation 4. Demonstrate the ability of reading and comprehending effectively 5. Solve problems through effective interpersonal interactions and team building					
Syllabus Outline					
• Introduction to communication and communication skills • Presentation skills • Listening and Observation skills • Effective reading • Report writing • Effective note taking • Interpersonal interactions and team building					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
3	MT2211	Mechanical Behaviour of Materials		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
3.0	2.0	4.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Explain the theoretical strength of materials and their deviations from actual strength based on structure 2. Identify main factors that influence the fracture stress and failure mode of materials 3. Recognize the key aspects in theories of material fracture, and their applications and limitations in designing of engineering components. 4. Design components for failure avoidance and selection of safety factors based on fracture mechanics. 5. Discuss the methods used in high speed materials characterization and mechanism in material failure at high strain rate. 6. Discuss mechanism of fatigue failure and identify causes that lead to fatigue failure. 7. Apply Goodman - Gerber diagram, Basic estimation of fatigue life, Minor's law of Cumulative fatigue for designing against fatigue failure. 8. Describe mechanisms of creep and predict creep-life of components under steady-state conditions for design applications. 9. Describe fracture mechanisms of polymers and ceramics and identify their main distinguishing factors.					
Syllabus Outline					
• Structure and deformation in Materials Concept of stress, Elastic deformation and theoretical strength, Introduction to Elasticity / Plasticity / inelastic deformation. • Stress–Strain Relationships Models for Deformation Behaviour, Elastic Deformation, St – Venants Principle, Three-Dimensional States of Stress, Yield criteria, Tresca criterion, Von mises criterion, Statically – indeterminate problems. • Mechanics of Materials Bending of beams, Torsion of circular bars, Principle Stresses (2D-transformation), Deflection of beams, Buckling of Columns/Struts. • High Strain Rate / Dynamic Behaviour of Materials Elastic stress wave propagation, Material Characterization at high strain rate, Material failure at high strain rate. • Fracture mechanics Stress concentration, Stress intensity factor, Griffith's criterion, Statistical Approach to Brittle Fracture and Safe Designs, Application of Fracture toughness to Design, Fracture Toughness Testing. • Fatigue Failure					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
3	MT2220	Ferrous Metals and Alloys		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
7/2	1.0	4.0	None	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Describe the basic principles underlying production of steel and cast iron. 2. Identify the phases in the microstructure of plain carbon steels and cast irons. 3. Explain the relationship between microstructure and properties of plain carbon steel and cast iron. 4. Apply the knowledge on structure and properties to select suitable plain carbon steels, alloy steels, cast irons and alloy cast irons for engineering applications. 5. Explain the basic heat treatment methods of steels and cast irons. 6. Define basics of heat treatment processes and their uses. 7. Describe the TTT and CCT diagrams and use them to explain microstructures and properties after heat treatments. 8. Identify the effects of heat treatment processes on the mechanical properties of different metals and alloys. 9. Design profiles of mechanical properties for engineering components considering its functional requirements. 10. Propose alternative designs to avoid failures.					
Syllabus Outline					
• The sources of metals • Ore preparation: Comminution; Sizing (screening and classification); Concentration • Pretreatment Processes Drying; Calcination; Roasting; Agglomeration • Thermodynamic evaluations of extractive metallurgical processes • Pyrometallurgy: Drying; calcination; roasting; oxide reduction; smelting; matte converting; fire refining; distillation • Hydrometallurgy: Leaching; precipitation techniques; ion exchange; solvent extraction; reverse osmosis • Electrometallurgy: Electrowinning and electrorefining • Extraction procedures for ferrous and non-ferrous metals: Blast furnace and alternative iron making processes; extraction and refining of Cu, Al, Zn, Au, Ti, Ag and Pb.					

Intake	2020 onwards	Specialisation	Electronic and Telecommunication Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
4	EN2853	Embedded Systems and Applications		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
2	2	3	None	60	40
Learning Outcomes					
1. Explain and analyse the basic building blocks of embedded systems 2. Explain design abstraction levels and corresponding design methodologies 3. Implement interfaces to integrate various sensors and actuators to an embedded system 4. Apply basic hardware and software design techniques to implement an industry-compatible embedded system					
Syllabus Outline					
1. Introduction to Embedded Systems Components and characteristics of embedded systems. Major challenges (low power, design complexity, verification complexity, time-to-market, security, and reliability). Design abstraction levels. Requirement analysis. Embedded design flow. 2. Embedded Computing Introduction to microcontrollers. Components of a microcontroller. Programming microcontrollers. Introduction to embedded processors and operating systems (ARM Core). Simulation software. Hardware-software co-design techniques. Inter-component communication. 3. Sensors and Actuators Sensors. Sensor interfacing. Actuators. A/D and D/A converters. Communication. 4. PCB Design Electronic circuit design software. Schematic design and component libraries. PCB stack-up, footprints, design practices, design rules, layout design, and design for manufacturability. Fabrication outputs. PCB manufacturing techniques. 5. System Deployment and Operation Enclosures. Noise and other environmental considerations. Reliability and stability					

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
4	MA2034	Linear Algebra	2	E	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
2	0	MA1014	30	70	
Learning Outcomes					
After the successful completion of this course students should be able to • Understand the concepts of vector spaces • Determine the linear transformation of a matrix, diagonalization of a matrix and inner product and norms on vector spaces • Apply Gram-Schmidt method for orthogonalizing the set of vectors in an inner product space					
Syllabus Outline					
• Groups and Fields. • Vector Spaces, Subspaces, linear combinations, span of a set, linear independence, bases, dimension • Inner product spaces, Orthogonal complements, Gram-Schmidt orthonormalization • Linear transformations, nullity, kernel, matrix form of a linear transformation, column space and row space of a matrix. • Special Linear Transformations • Dual space • Eigen values and eigenvectors, characteristic polynomial, Cayley-Hamilton theorem, minimal polynomial, diagonalization of matrices, QR factorization, quadratic forms. • Norms, spectral radius, functions of matrices • Positive/negative definiteness					

Semester	Code	Module Title	Credits	C/E/O	GPA/NGPA
4	MA3014	Applied Statistics	2	E	GPA
Hours/Week		Pre-requisites/Co-requisites	Evaluation (%)		
Lecture	Tute/Lab		CA	WE	
2	0	MA1024	30	70	
Learning Outcomes					
After the successful completion of this course students should be able to • Calculate a range of statistics and summarize the properties of statistical distributions related to the analysis of data • Analyze data and interpret results to derive conclusions by choosing appropriate statistical tests • Evaluate the real-world problems using probability, statistics and statistical modelling					
Syllabus Outline					
• Discrete distributions (Negative Binomial, Geometric, Hypergeometric) • Continuous distributions (Exponential, Gamma, Chi-Square, Fisher’s F-distribution) • Confidence intervals for means, proportion(s) and variance(s) • Hypothesis testing on means, proportions and variances • Contingency tables, Chi-square test of association and Goodness-of-fit test of distributions. • Simple linear regression • Introduction to sampling techniques					

Intake:**2020 onwards****Specialization:****Mechanical Engineering**

Semester	Code	Module Title	C/E/O	GPA / NGPA
4	ME2833	Mechanics of Machines	C/E	GPA

Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tute			CA	WE
3/2	2/2	2	ME1033	30	70

Learning Outcomes

After completing this module, the students should be able to:

LO-1: Demonstrate knowledge of principles in Mechanics of Machines

LO-2: Solve simple problems in Engineering Mechanics

Syllabus Outline**Section 1: Planar Linkages**

Velocity diagrams; Acceleration diagrams; Linkages involving sliding connections; Coriolis acceleration.

Section 2: Gear Drives

Geometry of spur gearing; External and internal gearing; Epicyclic gear trains; Speed – torque relationships.

Section 3: Balancing of Rotating Masses

Static and dynamic balancing of shaft-rotor systems.

Section 4: Single Degree of Freedom Vibrations

Response of viscous damped systems to harmonic excitation or ground vibration; Dynamic magnification; Transmissibility.

Intake:**2020 onwards****Specialization:****Mechanical Engineering**

Semester	Code	Module Title	C/E/O	GPA / NGPA
4	ME2851	Fundamentals of Machine Elements Design	C/E	GPA

Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tute			CA	WE
2	2	3	ME1033	30	70

Learning Outcomes

After completing this module, students should be able to:

LO-1: Define key terms in machine design and explain the design process

LO-2: Analyse stresses in mechanical components under different loading conditions

LO-3: Design and/or select machine elements according to the design criteria considering standards

Syllabus Outline**Section 1: Introduction**

Machine design & machine elements; Design process; Basic design considerations; Design standards.

Section 2: Loads and Stresses

Flow of forces through simple machine elements and stresses; Stress concentration; Common causes of failure of machine elements.

Section 3: Limits, Fits, and Tolerances

Limits; Tolerances: dimensional and geometric, standards in applying tolerances; Fits: clearance, transition, and interference; Hole basis and shaft basis systems of fits; Interpretation of engineering drawings with limits and tolerances.

Section 4: Power Transmission through Machine Elements

Shafts: types and terminology, design of shafts; Flywheels; Belts and belt drives: types and terminology, tensioning, selection of belts and pulleys; Gears and gear drives: types and terminology, tooth profiles, system of gears, design considerations, beam strength.

Section 5: Coupling Elements

Couplings: types and terminology, design of couplings; Keys & splines: types and terminology, standard dimensions, stresses in keys & splines, design of keys & splines.

Section 6: Fasteners and Joints

Screwed joints: definitions and terminology, types of nuts and bolts, standard dimensions and forces, design of screwed joints; Riveted joints: types and terminology, design of riveted joints; Welded joints: types and terminology, design of welded joints; Other fastening methods.

Section 7: Bearing Elements

Bearings: terminology and classification, rolling contact bearings, sliding contact bearings, bearing life and selection of bearings.

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
4	MT2171	Ceramic Engineering		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
7/2	1.0	4.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Identify the importance of powder properties, characterization, consolidation and forming in ceramic fabrication. 2. Describe the engineering processes of fabrication, drying, sintering and densification of ceramic products. 3. Identify causes of defects in ceramic products. 4. Design body mixtures and fabrication processes of clay-based ceramics, glass and glass ceramics. 5. Describe technology of bio ceramics, refractories, electro-ceramics and advanced structural ceramics.					
Syllabus Outline					
• Introduction to ceramics: Structure, properties and applications of traditional ceramics, advanced ceramics, glass and cement. • Ceramic powder preparation: Mechanical methods - comminutions, mechano-chemical synthesis. Chemical methods - solid state reactions, liquid solutions, vapour phase reactions. • Powder characterization: Particle size, particle size distribution, particle shape, surface area • Rheology and colloids: General introduction to colloids and rheology, isomorphous substitution, double layer theory. • Powder consolidation and forming of ceramics: Powder mixing, types of mixtures, powder consolidation methods, powder pressing techniques, Special ceramic fabrication techniques. • Design and Fabrication of clay based traditional ceramics: Plastic and non-plastic raw materials, preparation of body mixtures, forming methods, casting, extrusion, fabrication of selected ceramic products. • Drying and sintering: Shrinkage and swelling, sintering fundamentals, solid state sintering, viscous sintering, grain growth and microstructural evolution, pressure sintering (hot pressing), microwave sintering, reaction sintering, evaluation of sintered ceramics, finishing operations. • Design and formulation of Ceramic glazes and colour pigments. • Design and formulation of Glass and glass ceramics. • Introduction to Refractories, electro ceramics, bio ceramics and advanced structural ceramics.					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
4	MT2181	Solid State Materials		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
7/2	1.0	4.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Describe and apply the concepts of wave function, probability density and operators associated with quantum mechanics. 2. Solve the Schrodinger's equation for time independent potential functions. 3. Describe quantum mechanical explanation of atoms and solids. 4. Determine material properties using Drude model and outline the limitations associated with free electron gas models. 5. Identify the need for large system statistical mechanics and contrast between the Maxwell-Boltzmann Statistics and the Fermi-Dirac Statistics. 6. Calculate density of states using free electron model. 7. Explain the origin of bands through interaction of electron waves with Brillouin zones. 8. Correlate material properties using the band structure and density of state functions. 9. Write and utilize the equations for magnetic inductions and common magnetic parameters and distinguish magnetic materials based on susceptibility.					
Syllabus Outline					
Part 1: Engineering Quantum Mechanics • The wave function: The Schrodinger Equation; Normalization; Operators and Expectation values • Time-independent Schrodinger Equation: Solutions of Schrodinger equation for different potentials; Hydrogen atom, Angular momentum and Electron Spin • Identical Particles: Two particle systems; Atoms; Solids • Perturbation Theory: Zeeman Effect and Hyperfine Splitting in Hydrogen; Emission and absorption of radiation; Spontaneous emission. • The Variational Principle Part 2: Solid-state physics of Materials • Classical Model of solids: The free electron gas and the ideal gas; The Drude model • Statistical Mechanics: Maxwell-Boltzmann Statistics; Fermi-Dirac Statistics • Density of States: Electrons in solids; Particle in quantum well • Reciprocal Space and Brillouin Zones: Diffraction in reciprocal space; Interaction of electron waves with Brillouine zones • Bands: Free electron approximation; Band structure and types of materials • Quantum theory of Metals: Conduction in metals; The Seebeck effect and the thermocouple; Thermionic emission • Semiconductors: Intrinsic and extrinsic semiconductors; Temperature dependence; Optical absorption • Dielectrics: Matter polarization and relative permittivity; Polarization mechanisms; Gauss's law • Phonons: Harmonic oscillator and Lattice waves; Debye heat capacity • Magnetism: Common equations associated with magnetism; Susceptibility and distinguish between ferromagnetic, paramagnetic and diamagnetic; Magnetism and spin					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
4	MT2230	Kinetics of Materials		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Distinguish between instantaneous rates and average rates. 2. Determine the rate law from initial rate data and determine the order of reaction, the rate constant. 3. Describe the fundamentals of diffusion in solids. 4. Explain Kinetics of phase transformations.					
Syllabus Outline					
• Introduction to Kinetic Processes in Materials Introduction to reaction kinetics and transport process. • Reaction kinetics Concept of activation energy, kinetics of simple reactions, Kinetics of complex reactions, Reversible, competing, concurrent, and chain, Temperature Dependence of the Rate Constant • Chemical reactions of solids Decomposition of solid, Kinetics of complex reactions, solid + gas and solid + liquid, Solid-solid reactions • Diffusion in solid state Fundamentals of Diffusion, Random walk model, Fick's first and second laws of diffusion, Diffusion mechanisms, Diffusion with Finite Boundary Conditions. Application of diffusion (carburizing, doping etc.), Calculation of diffusion profiles • Fluxes, Forces, and interdiffusion Flux Density of Moving Particles, Mobility and Forces, Absolute Mobility, Relaxation Time, Stokes Law, Particle Settling, Diffusion in Liquids, Diffusion in molten metal at their Melting Points, Oxygen Diffusion, solid electrolytes. • Kinetics of phase transformations Solidification, Nucleation and Growth Kinetics, Solid/Solid Interface Migration Kinetics, Morphological Instability and Growth of Phases, Growth of Phases - Diffusion or Interface Reaction Control					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
4	MT3095	Polymer Engineering		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Describe the principles of rubber elasticity and explain Mooney-Rivlin theory for rubber-like materials for design calculations. 2. Choose appropriate viscoelastic models to design and analyse simple polymer engineering problems. 3. Identify and categorise rheological behaviours of polymer melts and analyse and design their flow characteristics in geometrical channels. 4. Discuss the significance of morphological manifestations of polymers in understanding the fundamental physical and chemical properties of polymer components 5. Identify the physical characteristics of polymers by stress-strain behaviour and relate them to the structure of the polymers and use this information in design of polymer components.					
Syllabus Outline					
● Rubber elasticity Chain conformations in polymers, statistical theory of rubber elasticity, entropy, internal energy, effect of temperature, theory of Mooney and Rivlin. ● Viscoelasticity Visco-elastic models, Stress-strain-time relationship, Superposition, retardation and relaxation time spectrums, WLF equation, Dynamic behaviours of polymers, Frequency-temperature equivalence problems. ● Rheology of polymers Newtonian and non-Newtonian fluids, Time dependent fluids, Time independent fluids, Flow curves, Isothermal channels flow, Boundary effect, Equipment used to obtain flow data ● Morphology of polymers Polymer crystallography, Lamella, Spherulites, effect of crystallinity, Orientation induced crystallization, Chain orientation, Properties of oriented polymers, Frozen stress, Stress concentration. ● Yield and fracture Behaviour of polymers under low strain rate and high strain rate loading conditions. Dependence of mechanical properties according to glass rubber transition temperature.					

Intake:**2020 onwards****Specialization:****Mechanical Engineering**

Semester	Code	Module Title	C/E/O	GPA / NGPA
5	ME3813	Machine Design	C/E	GPA

Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tute			CA	WE
1	2	2	ME1091, ME2851	100	0

Learning Outcomes

After completing this module, the students should be able to:

LO-1: Solve a comprehensive design task

LO-2: Make use of parametric solid modelling techniques in creating virtual documents

LO-3: Recommend design information through technical reports, engineering drawings, computer representations, and oral communication

Syllabus Outline**Section 1: Design Methodology**

Design models; Design methods.

Section 2: Problem Formulation

Consideration of loads; Context; Constraints.

Section 3: Conceptual Design

Concept generation techniques; Concept selection.

Section 4: Selection of Materials

Criteria: mechanical properties, cost, manufacturability, and application.

Section 5: Detailed Design Calculation

Load and stress analysis; Selection of prime mover; Selection of materials and machine elements.

Section 6: Solid Geometric Modelling

Principles; Tools and techniques of parametric solid modelling.

Section 7: Production Drawings

Presentation of details required for manufacturing.

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
5	MT2075	Metal Forming and Joining		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
3/2	1.0	2.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Describe the mechanical and metallurgical aspects of metal forming processes. 2. Explain the basic principles of metal forming processes. 3. Discuss the application of metal forming processes for engineering components. 4. Select and Design suitable joining techniques for metals and alloys 5. Describe basic principles of welding processes. 6. Describe mechanical properties and metallurgical aspects of welded joints. 7. Select suitable welding processes for metals and alloys in the context of manufacturing system requirements.					
Syllabus Outline					
Metal Forming • Classification of metal forming processes • Theories on metal forming Tool-work piece interaction; Homogeneous deformation; Inhomogeneous deformation; Friction, Tresca's criterion, Von Mises's criterion; Strain hardening, recovery, recrystallization and grain growth. • Direct compression type bulk forming processes Forging and rolling. • Indirect compression type bulk forming processes Wire drawing and extrusion. • Sheet metal forming processes Deep drawing; Shearing; bending etc. • Defects in metal forming processes Incomplete forging penetration; Surface cracking, cracking at the flash; Cold shut, cracking due to surface (secondary) tensile stresses. Joining • Introduction and classification of metal joining processes • Material, Metallurgical, Mechanical and Design aspects of joining • Types of weld joints and their applications • Classification of welding Processes Arc welding processes; Gas welding processes; Resistance welding processes; Solid state welding processes • Temperature distribution, change of microstructure and weld strength • Welding defects, weld inspection and testing					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
5	MT3031	Degradation and Failure of Materials		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	None	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Describe degradation and corrosion mechanisms of engineering materials. 2. Design protection systems against different forms of corrosion 3. Design polymer components for controlled degradation. 4. Identify, analyse and investigate material related failures. 5. Identify probable causes of failure and design against failures					
Syllabus Outline					
• Introduction to degradation Degradation principles, electrochemical aspects, Polarization, environmental effects, metallurgical aspects. • Corrosion Definition of corrosion, Cost of corrosion, Corrosion damage, Corrosion rate expressions, general Corrosion, galvanic corrosion, crevice corrosion, pitting corrosion, intergranular corrosion, selective leaching, erosion corrosion, stress corrosion, hydrogen damage. • Oxidation of Metals Oxidation principles, Types of oxidation, kinetics of oxidation • Degradation of ceramics Acid brick, stone ware and porcelain, structural clay, glass, concrete • Degradation of polymers • Thermal, mechanical, photo, bio and chemical degradation. Degradation due to radiation and hydrolysis. • Design of Corrosion Protection Methods Material selection, alteration of environment, design of cathodic and anodic protection, design of coatings systems, corrosion prevention of buried metal structures • Introduction to Failure Analysis Reasons for failure, Objectives of Failure analysis, Principles and Approaches in Failure Analysis Work • Steps of Failure Investigation • Negotiating Goals of Investigation The priority of the investigation, the resources available, Dealing with constraints, The goal of the investigation • Root cause and likelihood analysis Likelihood analysis, Probability of Occurrence, Convergence on root causes, Severity Effect, Corrective action Identification • Case Studies					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
5	MT3054	Characterization of Materials		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
7/2	1.0	4.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Describe characterization techniques to probe specific materials characteristics. 2. Define the scope of a specific characterization method. 3. Compare different complementary characterization methods. 4. Design an apparatus/technique to monitor material characteristics.					
Syllabus Outline					
• Powder X-ray Diffraction Diffraction by small crystals; Structure factor for Bragg reflections; Effect of temperature variation on the intensity; Diffractometer recording of powder patterns; indexing of powder patterns; phase identification, lattice parameter determination; use of space groups in structure determination (if time permits) • Vibration spectroscopy Absorption process; Molecular Stretching & Bending; IR spectrometers –a) Dispersive b) Fourier transform; Correlation charts & analysis of spectrum; Background spectrum & Baseline correction. • Quantitative aspect of spectro-chemical measurements Absorption method; Beer's law; Relaxation process • Atomic absorption & emission spectrometry Instrumentation; Interference in AA spectroscopy; Calibration curves and detection limits • Thermal analysis Principles of Differential Scanning Calorimetry; Principles of Thermo Gravimetric Analysis. • Optical microscopy Resolution and contrast in microscopy; Polarizing microscopy • Electron microscopy Interaction of electron beam with matter; Optical system of an electron microscope; SEM and TEM; EDAX and WDAX analysis					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
5	MT3060	Industrial Metallurgy		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Identify and classify the manufacturing processes related to metallic materials. 2. Explain general casting operation, and casting processes and casting defects 3. Explain the solidification theory of metals. 4. Explain powder metallurgy process including metal powder production techniques and powder compaction techniques 5. Discuss the application of machining processes for manufacturing of engineering components 6. Apply the knowledge on mechanical and metallurgical aspects to prevent the defects occurring during the machining. 7. Select suitable joining methods for materials in the context of manufacturing and system requirements. 8. Explain metal additive manufacturing process					
Syllabus Outline					
• Classification of manufacturing Processes • Casting processes Sand casting, Continuous casting, Investment casting, Centrifugal casting, Die casting (Pressure and Gravity), Shell mould casting, Expandable pattern casting, Ceramic mould casting, Plaster mould casting • Solidification theory Solidification of pure metals and alloys, grain formation, dendritic growth etc. Heat transfer during solidification • Melting and type of furnace • Casting design and material selection. Design of component for casting, Design of runners, risers, and gates., Avoidance of aspiration • Casting defects • Introduction to powder metallurgy. Application of P/M techniques and its advantages, properties. • Production and properties of metal powders. Atomization, reduction of metal oxides, electrolysis, pulverization, powder properties. • Compaction and sintering of metal powders • Mechanical properties of powder metallurgical Products, Applications Introduction to machining • Introduction to metal additive manufacturing • Classification of machining processes • Machinability • Machinable materials • Select suitable joining methods for materials in the context of manufacturing and system requirements. • Mechanical Joining, Adhesive bonding, soldering and brazing, joining of dissimilar and nano materials.					

Intake	2020 onwards	Specialisation	Materials science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
5	MT3084	Latex Science and Technology		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Describe the composition and constitution of natural rubber latex 2. Discuss the stabilization and destabilization mechanisms of natural rubber latex 3. Compare the latex concentration techniques 4. Describe the tests involved in latex characterization 5. Explain the functions of latex compounding ingredients and pre vulcanization process 6. Discuss technologies involved in different types of latex based products manufacture. 7. Propose suitable latex formulations for service requirements 8. Identify special features of synthetic lattices					
Syllabus Outline					
• Describe the composition and constitution of natural rubber latex Definition of latex, Classification of lattices ,Colloidal destabilisation of lattices, Composition and constitution of field natural rubber latex • Discuss the stabilization and destabilization mechanisms of natural rubber latex Spontaneous coagulation of latex, Preservation of latex • Compare the latex concentration techniques Techniques used for concentration of NR latex; Evaporation, Creaming, Centrifugation and Electro-deposition • Describe the tests involved in latex characterization Characterisation of concentrated NR latex- DRC, TSC, VFA, Alkalinity, KOH number, MST etc. • Explain the functions of latex compounding ingredients and pre vulcanization process Latex compounding ingredients - Vulcanising agents, Accelerators, Fillers, Stabilisers, Antioxidants , Surfactants etc. , Preparation of dispersions and emulsions, Pre vulcanisation process • Discuss different technologies involved in different types of latex based products manufacture Technologies involved in Dipped products, latex foam, latex thread, latex tube, latex paints, adhesives etc. • Propose suitable latex formulations for different service requirements Formulation design for latex based products such as Dipped products, latex foam, latex thread, latex tube, latex paints, adhesives etc. • Identify special features of synthetic lattices Industrially important synthetic lattices, special features and applications					

Intake:	2020 onwards	Specialization:					
Semester	Code	Module Title			Credits	C/E/O	GPA / NGPA
6	HM2450	Introduction to Psychology			2	E	GPA
Hours/Week		Prerequisites / Corequisites				Evaluation %	
Lecture	Lab/Tutes					CA	WE
2		None				100%	
Learning Outcomes							
After completing this module, students will be able to							
1	explain the basic principles and theories of human psychology and apply them to personal, organisation and social issues to understand, predict and modify human behaviour.						
2	discuss differences in individual mental processes and human behavior						
Syllabus Outline							
1	Goals of Psychology						
2	Evolution of Psychology and its theories						
3	Self-evaluation and life skill development						
4	Personality Development						
5	Introduction to Emotional Intelligence						
6	Different approaches of Psychology						
7	Building of Human Relationships						
8	Attitudes and decision making						
9	Stress Management						
10	Addiction						
11	Psychological Intervention						

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
6	MT3350	Materials Modelling		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
2.0	2.0	3.0	none	50	50
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Distinguish the modelling and simulation techniques applicable for materials in different time and length scales 2. Describe the general theoretical background behind materials modelling methods. 3. Develop an understanding of the assumptions and approximations that are involved in modelling frameworks. 4. Judge the applicability of the different theoretical calculation methods to solve specific materials problems. 5. Interpret the simulation results to make useful conclusions.					
Syllabus Outline					
• Introduction to Modeling and Simulation Definition of a model / mathematical model – key features of a mathematical model / Numerical Simulations Modeling and simulation in Materials Science and Engineering – Time scales and length scales • Continuum Scale Modeling (Finite Element Analysis) Different continuum scale modeling approaches. Theory and fundamentals of FEA Analysis. Error, Error Estimation and Convergence. Computer Implementation and Modeling Consideration of FEA. Introduction to simulation software Simulation of practical problems • Molecular Modeling (Molecular Dynamic (MD) Simulation) Theory of classical molecular dynamic simulation, potential energy functions, integration algorithms MD simulation software Equilibrium MD simulation Applications of MD simulation • Density Functional Theory (DFT) Introduction to Density Function Theory; Background, Exchange correlation functionals Plane wave based DFT calculations; Basis sets, Pseudopotentials Equilibrium properties and surfaces from DFT calculations; Surface structures, Surface relaxation, Surface reconstruction, Surface energy Applications of density functional theory • Advantages and limitations of simulation techniques					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
6	MT3880	Engineer and Society		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tute			CA	WE
1	4	3	None	100	0
Learning Outcomes					
After completing this module students will be able to: 1. Demonstrate an understanding of the responsibilities of the engineering profession and its social context 2. Demonstrate an understanding of the health, safety and environmental requirements of the society 3. Practise with integrity in the social context of the engineering profession with an understanding of ethical issues 4. Identify and apply appropriate tools/ techniques for the evaluation of health, safety and environmental hazards/ consequences and risk assessment 5. Interpret the engineers' role in ethically assuring healthy, safe and excellent environmental conditions targeting the overall sustainable development of the society 6. Ability to critique technology 7. Apply the knowledge and skills gained of towards building character as a socially responsible professional engineer.					
Syllabus Outline					
1. Introduction to Engineering Ethics - Historical context, moral responsibility, IESL code of ethics, other relevant codes of ethics, community standards and personal responsibility, ethics of research and publication 2. Ethics in the Society & Workplace - Respect for social & cultural values, respect for other professions, social responsibility, ethical decisions as individuals, identifying ethical issues, conflicting scenarios and problems in the field of engineering, leading organizations towards ethical behaviour 3. Inclusive engineering concepts – ensuring that engineering products and services are accessible and inclusive of all users, and are as free as possible from discrimination and bias 4. Legal requirements related to engineering practice – acts, ordinances and regulations 5. Health & Safety – Definitions, areas and hazard identification, risk assessment, evaluation and management 6. Health & Safety Management – Management practices, local regulations, global standard and best practices, designing of health and safety management systems, special topics 7. Environment – managing the generation, transportation and disposal of waste in industry, overview of controlling and treatment technologies, local standards and EPL procedure, introduction to environmental impact assessment 8. Ethical issues in emerging technologies 9. Engineers' responsibility in sustainable development 10. Case studies (industry specific)					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
6	MT3993	Industrial Training		C	NGPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
-	-	6.0		100	0
Learning Outcomes					
At the completion of this module, students should be able to; 1. Describe the fundamentals of materials science and engineering in industrial processes. 2. Identify, formulate and model industrial problems and propose engineering solutions based on a systematic approach. 3. Recognize the social, cultural, global, health, safety and environmental responsibility as an engineer. 4. Appreciate managerial practices within the organization					
Syllabus Outline					
Students will undergo 24 weeks of industrial training in one or more of organization of the following categories, • Industries in Material processing • Industries in Materials related manufacturing • Material testing or standardizing organizations • Research and Development organizations • Industries in Materials Engineering Allied Fields					

Intake	2020 onwards	Specialisation	Faculty of Business (Dept. of Industrial Management [IM])		
Semester	Code	Module Title	Credits	C/E/O	GPA / NGPA
7	MN4023	Engineering Economics	2	C/E/O	GPA
Hours/Week		Prerequisites / Corequisites	Evaluation %		
Lecture	Lab/Tutes		CA	WE	
2	-	None	30	70	
Learning Outcomes					
LO1	Identify the most relevant economic concepts for the engineering decisions				
LO2	Apply these concepts to practical engineering projects and decisions				
Syllabus Outline					
Fundamentals; time value of money, equivalence and cash flow diagrams (2 hrs)					
Discounted cash flow; time value equivalence, single payment and annuity factors and numerical examples. Cash flows and compounding; (4 hrs)					
Comparison methods; assumptions, net present value, annual worth, equivalent annual cost with/without salvage value, equivalent annual worth of fixed asset lives and perpetual lives, internal rate of return (IRR) and minimum acceptable rate of return and IRR irregularities, numerical examples; (6 hrs)					
Analysis of alternatives; classification, mutually exclusive alternatives, incremental analysis and preferred method for decision making; (3 hrs)					
Project feasibility analysis; financial feasibility, market price analysis, cost of capital and weighted average, economy feasibility, shadow pricing, benefit cost (B/C) analysis, irregularities of B/C analysis and preferred method for decision making; (4 hrs)					
Sensitivity analysis and decision trees; What if, sensitivity graph and interpretation of the analysis, discounted decision trees and application of decision trees; (2 hrs)					
Risk management; Risk identification, risk analysis and risk response; (2 hrs)					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
7	MT3201	Comprehensive Design Project		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
-	8.0	4.0	none	100	0
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Apply the fundamental components or steps in the engineering design process. 2. Work collaboratively with others and articulate practices that lead to successful teamwork. 3. Develop an engineering design proposal. 4. Apply theoretical concepts in designing. 5. Demonstrate an understanding of the importance of ethics, professionalism, globalization, safety and sustainability in design and manufacturing. 6. Apply appropriate planning strategies to solve open ended design problems. 7. Produce project records and project schedules. 8. Recognize the use of appropriate standards and specifications.					
Syllabus Outline					
• Requirement Analysis (Society / Industry) - Types of requirements - Eliciting requirements - Analysing requirements - Recording requirements • Design of products - Physical (Mechanical, Thermal, Optical) analysis - Formulation of specifications (material, microstructure, physical and mechanical properties) - Presentation of Design proposal • Design and selection of manufacturing process - Raw material selection - Selection of processing method - Selection of machines and tools, - Design of customized machines/tools (dies, moulds, etc) • Design of quality assurance procedures - Testing requirements as per product standard - Selection of testing instruments - Design of testing schedule. • Presentation of design - Report - Oral presentation - Machine drawings					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
7	MT4024	Total Quality Management		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
3/2	1.0	2.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Explain the importance of total quality management concepts, tools and promotional vehicles and its applications. 2. Design and implement quality programs according to the industrial requirements. 3. Apply total quality management concepts for achieving organizational goals maximizing customer satisfaction.					
Syllabus Outline					
• Benefits of quality to individuals, organizations, customers, suppliers, and society. • Key events in the evolution of quality. • Key components of total quality management and how organizations approach total quality management deployment. - Total quality management concepts - Total quality management tools - Promotional vehicles (daily work management/policy work management) • Role of process management in total quality management and the interrelationship of processes and systems in service & Manufacturing organizations. • Basic quality tools used to help improve processes/ problem solving. - Quality circles - Cross functional teams/quality improvement teams • Define different quality concepts related to quality evolution, total quality management, process management, and the basic quality tools. • Identify the national and international total quality management Promotions & Evaluations - NCQP/NPA/NQA/Deming award/ ICQCC					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
7	MT4074	Design and Fabrication of Polymer Products		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Identify the product requirement and design criteria for plastic and rubber items. 2. Explain the design concepts and engineering considerations in designing plastic and rubber items. 3. Apply fundamental knowledge to design extruded and moulded items of rubber and plastic to given specifications. 4. Recognize the importance of maintaining appropriate process parameters to achieve design specifications. 5. Analyse defects in plastic and rubber items to find appropriate remedies.					
Syllabus Outline					
• Design of and rubber products (Moulded and extruded) Product design examples: Static and dynamic rubber springs, design of tyres, hoses and seals • Design of plastic products (Moulded and extruded) Product design examples; blow moulded products, Injection moulded products, blowing film design, plastic hollow shape designs, gear and bearing design • Design Troubleshooting Design related, material related and process related.					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
7	MT4130	Selection of Materials for Engineering Applications		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	None	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Describe different criteria/aspects in material selection 2. Identify different material selection processes 3. Use material property charts/databases in material selection 4. Describe Techno-economic aspects of materials selection 5. Apply the appropriate material selection processes for different engineering applications.					
Syllabus Outline					
• Material property charts ○ Importance of material property charts in material selection ○ Relationship of material properties through material property charts. • Materials selection process ○ Materials selection strategy -overview ○ Material selection through material indices / property charts ○ Techno-economic aspects of materials selection ○ Decision matrices in materials selection ○ Selection of materials for multiple constraints, and conflicting objectives. ○ Computer-aided materials selection • Shape efficiency factor-based material selection ○ Introduction to shape factor of a material ○ Material indices with shape factor ○ Materials architecture at micro level ○ Case studies in shape efficiency-based materials selection. • Application of Materials selection processes for Industrial Aspects ○ Requirement pyramid ○ Product character ○ Creation of product personality • Case studies in materials selection for engineering applications.					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
7	MT4810	Continuum Scale Numerical Simulation of Material Systems		E	GPA
Hours/Week		Credits	Prerequisites / Requisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Define the numerical simulation process in continuum scale in modelling engineering materials. 2. Describe the process in applied finite element analysis with a commercial software. 3. Apply numerical simulation procedures of engineering materials in industry and research domains. 4. Critique the validity of numerical simulation based results through experimental techniques.					
Syllabus Outline					
• Introduction to Mathematical Modelling. • Experimental Characterization Techniques for Material data. • Numerical Simulation of Physical Systems. • Introduction to Applied Finite Element Analysis. • Direct Stiffness Method and Stiffness Matrix. • Shape Functions, Iso-parametric Elements and Stiffness Matrix Integration. • Introduction to commercial numerical simulation software. • Introduction to Continuum Mechanics Foundations. • Numerical Simulation of Mechanical and Thermal Behaviour of engineering and bio-materials. • Numerical Simulation of Fracture of Engineering Materials.					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semesters	Code	Module Title		C/E/O	GPA / NGPA
6,7,8	MT4204	Research Project		C	GPA
Hours/Week		Credits	Prerequisites / Co-requisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
-	14.0	7.0	None	100	0
Learning Outcomes					
At the completion of this module, Students should be able to;					
1. Utilize research skills to identify and analyse engineering problems. 2. Develop a research question, hypothesis or design related to materials science and engineering. 3. Conduct a critically focused literature review. 4. Design and evaluate methodologies to solve engineering research problems. 5. Develop competence in experimental and analytical techniques. 6. Use a range of appropriate data collection methods. 7. Analyse data according to accepted models of analysis, showing awareness of alternative models of analysis and theoretical frameworks. 8. Practice scientific writing and participate in the peer review process. 9. Express the relevance and significance of the outcomes/ conclusions of the research project by written and oral communication. 10. Comprehend and follow laboratory safety procedures and follow good laboratory practices.					
Syllabus Outline					
• Introduction to Research Methodology Basic procedures in research methodology throughout the project tenure. Basics of Good Laboratory Practice (GLP) and laboratory safety procedures. • Identification of the Research Needs Significance of the research project. Viability of the research project. • Literature Survey: Evaluate current literature relevant to the topic Demonstrate the quality of the own research. Provide additional information or background material. Place the project into the context of established work in the field. • Finalizing the preliminary research and submission of progress reports The student shall submit a report on his/her findings on the preliminary investigation (literature survey) and should devise a plan to carry out the research further. on the approval of the supervisor, student shall proceed with the experimental part of the research. It is necessary for the student to hold weekly meetings with the supervisor. • Research project report and the final appraisal The student should submit a draft report on the finding on or before the stipulated date. Student shall present findings before a panel consisting of academic staff members. Student research/analytical skills, communication skills and oral presentation skills will be evaluated.					

Intake	2020	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
8	MT4420	Energy Materials		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to;					
1. Describe working mechanism and performance characteristics of selected energy harvesting, energy storage and energy transmission devices 2. Relate material behaviour to performance characteristics of selected energy-related devices. 3. Appraise the critical role of advanced materials for high-performance and highly efficient energy harvesting, storage and transmission technologies.					
Syllabus Outline					
• Introduction to Energy Materials Overview on photovoltaics, Li- ion batteries, supercapacitors, and superconductors • Materials for Energy Harvesting Fundamental properties of semiconductors including crystal structure, Energy bands, Density of status, pn- junction diode electrostatics, carrier transport, light absorption, and recombination. Photovoltaic materials specially silicon, physical properties, and manufacturing process. Solar cell fundamentals including generation rate, I-V characteristics, and spectral response. Very- high efficiency concepts specially multijunction solar cells; spectrum splitting, cell configurations • Materials for Energy Storage Electrochemical energy storage fundamentals; Configuration and working mechanism of Li-ion secondary batteries and supercapacitors; performance characteristics Active materials used for anode/cathode in Li-ion secondary batteries; Nanomaterials for high-performance Li-ion batteries, synthesis processes, properties and performance Carbon-based and graphene-based electrode materials used for supercapacitors, mesoporous and nanoporous electrode materials for high-performance supercapacitors, synthesis processes, properties and performance • Superconducting materials for Energy Transmission Introduction to superconductivity as an energy saving concept: zero resistivity Basic principles and theories on superconductivity: Critical temperature, critical magnetic field and critical current density; Diamagnetic behaviour of superconductors: Meissner effect, Levitation effect; Ginzburg-Landau theory, BCS Theory, London Theory, Josephson tunnelling; flux pinning effect Superconductor Classification based on different criteria Applications of superconductors: High performance smart grid, Electric Power transmission cables, Transformers, Electric motors, Magnetic levitation Trains, MRI/NMR Machines					

Intake	2020 onwards	Specialisation	Faculty of Business (Dept. of Management of Technology [MOT])		
Semester	Code	Module Title	Credits	C/E/O	GPA / NGPA
8	MN4043	Technology Management	2	C/E/O	GPA
Hours/Week		Prerequisites / Corequisites	Evaluation %		
Lecture	Lab/Tutes		CA	WE	
2	-	None	30	70	
Learning Outcomes					
LO1	Recognize basic concepts and theories of management of technology				
LO2	Identify the usage of technology management concepts and theories in modern organizations and economy				
Syllabus Outline					
Strategic management of technology (6 hours)					
Technology-strategy relationship					
Elements of technology strategy and formulation of a technology strategy					
Integration of technology strategy and business strategy for competitive success					
Technology, the environment and sustainable development					
Acquiring technology through research and development (3 hours)					
The concepts of invention and innovation					
Definition and classifications of research and development					
New product development					
Challenges in commercializing research results					
Organizational Aspects of technology management (6 hours)					
Human dimension of technology and concepts of the entrepreneur and entrepreneur					
Organizational cultures and structures for promotion of creativity and innovation					
The imperative of knowledge management					
The learning organization					
Acquiring technology through technology transfer (3 hours)					
Motivations for acquiring technology through technology transfer					
Elements of technology transfer process					
Success and failure factors in technology transfer					
National innovation systems for facilitating technology-based development (4 hours)					
Concepts of the national innovation system (NIS) and science and technology infrastructure					
Comparison of NISs of developed, developing and first and second tier NIC countries					
State involvement and growth of science and technology parks in developed and developing countries					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
8	MT4121	Cleaner Production		C	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Identify causes for resource depletion and environmental pollution during industrial production. 2. Outline historical timeline and sustainability approaches that lead to current cleaner production (CP) concept. 3. Explain cleaner production concept, principles, methodology and initial tasks for cleaner production 4. Analyse processes using CP assessment methodology, construct process flow diagrams, Quantify resource flows, material balance and data collection in CP assessments. 5. Identify causes for waste generation, costing of wastes and apply CP options in CP assessment. 6. Recognise the importance of life cycle analysis for sustainable manufacturing and applying green supply chain management for industrial manufacturing processes. 7. Compare characteristics of industrial waste and consumer waste, waste segregation and separation methods used in industrial processes 8. Apply Eco- design principles and practices for sustainable manufacturing processes to enable climate change mitigation at industrial level 9. Select international standards and certification related to environment performance in manufacturing technologies to produce environmentally friendly products 10. Explain green chemistry and green engineering principles and their applications 11. Identify smart chemicals management practices for industries 12. Explain sustainable consumption and Production – tools and Strategies and sustainable Development Goals and programmes. 13. Describe circular economy – concept and application techniques. 14. Explain importance of applying Eco-innovation for business sustainability and constructing energy efficient buildings.					
Syllabus Outline					
● Resource depletion and environmental degradation ● Historical timeline and sustainability approaches ● Cleaner production concept, principles and methodology and initial tasks for cleaner production assessment ● Process analysis using CP assessment methodology, drawing the process flow diagram ● Quantification of resource flows, material balance and data collection in CP assessments ● Waste cause analysis and option generation in CP assessment techniques such as Ishikawa diagram, brain storming etc., ● Costing of wastes and identifying CP potential ● Feasibility analysis of CP options and option prioritization (weighted sum matrix analysis) ● Life cycle analysis and CP/defining system boundaries (scope & goal setting, inventory analysis, impact analysis) ● Characteristics of industrial waste and comparison with consumer waste, waste segregation and separation methods used in industrial processes					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
8	MT4084	Dies and Moulds for Polymer Processing		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Identify the product requirement and design criteria to design moulds and dies for plastic and rubber items. 2. Select appropriate materials, apply suitable hardening and surface treatments, choose correct Forming and Machining Techniques and Designing of heating and cooling systems for moulds and dies for polymer products. 3. Explain the design concepts and engineering considerations in designing of compression and transfer moulds according to industrial requirements. 4. Design blow moulds, injection moulds and extrusion dies for polymer manufacturing industry according to product specifications. 5. Identify the causes of defects in moulded and extruded products and suggest appropriate corrective measures.					
Syllabus Outline					
• Introduction to processing tools General Mould and Die design Criteria, Selection of Mould and Die Materials, Hardening and surface treatment methods, Forming and Machining Techniques, Designing of heating and cooling and systems. • Compression and Transfer mould Design Types, Principles. Design of pot and plunger systems, Determination of possible number of cavities, design of mould cavities, Design of loading chamber and feed system. Design of moulds for rubber items. • Blow Mould Design Mould Cooling Systems, Mould Venting, Pinch off, Parison Diameter Calibration, Wall thickness control, Blow Ratio Control, Design Considerations for Shrinkage, Neck, Shoulder, Threads and Bottom Section Design • Extrusion Die Design Characteristics of polymer melt, Die Geometry and Die head Pressure, Die Land Design, Extruder Die Swell, Classification of Dies- Dies for solid rod extrusion, Dies for hollow profiles, Blown film dies, Flat film dies Wire and cable coating dies, Spiral mandrel die, Fish tail die and Adjustable core die. • Injection mould design Cavity design, Layout of cavities in multi impression moulds feed system Design of Runners and Gates, Ejection systems, Cooling Systems, Venting system, other aspects					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
8	MT4301	Non-Destructive Testing		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
3/2	1.0	2.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Compare the relative advantages and drawbacks of the available Non-destructive testing (NDT) techniques 2. Choose the appropriate NDT technique with respect to the material and potential flaws 3. Develop NDT procedures in accordance with the relevant standards 4. Apply the relevant NDT procedures in a laboratory and on site.					
Syllabus Outline					
• Physical principles of basic NDT methods • Application of basic NDT methods • Comparison of basic NDT methods • Selection of NDT techniques for practical applications • Developing NDT procedures with relevant to the appropriate standards • Specific applications; NDT of welds and castings • Advanced NDT techniques					

Intake	2020 onwards	Specialisation	Materials Science and Engineering		
Semester	Code	Module Title		C/E/O	GPA / NGPA
8	MT4430	Compounding and Testing of Polymers		E	GPA
Hours/Week		Credits	Prerequisites / Corequisites	Evaluation %	
Lecture	Lab/Tutes			CA	WE
5/2	1.0	3.0	none	40	60
Learning Outcomes					
At the completion of this module, students should be able to; 1. Explain theories of mixing of solids, liquids and liquid solid mixes 2. Analyse and modify polymer formulations and blends. 3. Design polymer formulations and select appropriate testing procedures 4. Select and apply suitable testing procedures to assess performance and lifetime of polymer products 5. Analyse polymers to determine conformity of properties to physical requirements					
Syllabus Outline					
• Introduction to science of mixing Mixing entropy measures and its applications, mixing indices, scale of segregation and intensity of segregation, kinetics of mixing, rheology of filled polymers, overview of polymer mixing and blending machinery. • Ingredients in Mixing, blending and compounding polymer blends and formulations, filled polymers and polymer composites, compounding practice, mixing types, solid additives, morphology of filler additives, filler reinforcement, compatibility enhancers, filler surface modification and interfacial agents, dispersion of polymer nanoparticles in polymer, melts. • Applications of mixing and compounding Inorganic filler compounding, glass fibre compounding, Nano-composite compounding, reactive extrusion, recycling applications, reactive compounding and blending, elastomer compounding (NR, SBR, BR, IR, EPDM etc.), compounding materials used for unsaturated polyester, PVC mixing and compounding, compounding practices for other polymers, compounding for cable and profile extrusions, compounding economics. • Physical testing of polymers Theories of testing , significance in procedure and interpretation of results in tensile testing, hardness testing, measurement of softening temperatures, Impact testing, abrasion testing, Flexural testing, Tear resistance testing, fatigue testing, Rheological testing, Dynamic mechanical analysis, ageing tests, Thermal conductivity, Volume and surface resistivity, Dielectric strength, Flammability					

Intake:	2020 onwards	Specialization:	Languages				
Semester	Code	Module Title			Credits	C/E/O	GPA / NGPA
8	HM2030	Japanese as a Foreign Language			2	E	GPA
Hours/Week		Prerequisites / Corequisites				Evaluation %	
Lecture	Lab/Tutes					CA	WE
2	n/a	None				100	0
Learning Outcomes							
After completing this module, students will be able to							
1	Identify katakana characters and simple kanji characters.						
2	Practise a simple self-introduction in Japanese.						
3	Practise to count up to 100 and express date and time in Japanese.						
4	Write and read hiragana characters						
5	Apply simple Japanese expressions used in daily life to produce an effective dialogue						
6	Translate simple Japanese sentences into English.						
7	Compose simple sentences using basic grammar						
8	Develop student awareness of the general social and cultural background of Japan						
Syllabus Outline							
1	Introduce Katakana and Kanji characters, writing rules and words						
2	Starting with the Name, Age, Nationality etc., countries/interests/family, occupations and use of greetings						
3	Numbers (0-100), Practise saying date and time						
4	Introduce Hiragana characters and writing rules						
5	Invitation sentences, describing the location of people and objects, expressing opinion, making requests, reporting speech, conversation practice and role plays						
6	Use excerpts from the recommended texts for translation						
7	Basic grammar points, adjectives, adverbs, conjunctions						
8	Introduce foods, customs, festivals, education and new cultural trends in Japan						

Intake: 2020 onwards

Specialization: Materials Science and Engineering

Semester	Code	Module Title	Credits	C/ E/ O	GPA / NGPA	Hours/Week		Prerequisites / Corequisites	Evaluation %	
						Lecture	Lab/ Tutes		CA	WE
8	MT4120	Biomaterials and Medical Implants	2.0	E	GPA	2	-	None	20*	80

* CA comprises a Group Presentation.

Learning Outcomes

After completing this module, students should be able to:

1. Identify and describe the structural and functional aspects of essential biomolecules (carbohydrates, lipids, proteins, and nucleic acids).
2. Describe bone histology, the mechanical properties of bones, and the biological mechanisms of fracture healing.
3. Evaluate the suitability of metallic alloys (e.g., 316L SS, Co-Cr-Mo, Ti alloys) for orthopedic implants, considering mechanical properties, corrosion resistance, and metal hypersensitivity.
4. Explain the interaction of biomaterials with the human immune system, detailing the processes of inflammation, osseointegration, and wound healing.
5. Select and justify appropriate biomaterials for highly specialized applications including dental, ophthalmic, vascular, and auditory systems.
6. Assess various surface modification techniques (physical, chemical, and biological) aimed at improving biocompatibility and reducing implant failure.

Syllabus Outline

1. **Biomolecules:** Carbohydrates, lipids, amino acids, proteins, nucleic acids, and their biochemical importance.
2. **Bone Histology & Biomechanics:** Bone anatomy, cellular components, mechanical properties, types of fractures, and the process of bone repair using biomaterials.
3. **Metals and Alloys in Implants:** Mechanical properties, corrosion mechanisms, and toxicities of common metallic implant materials (SS, Co-Cr, Ti).
4. **Interaction with the Immune System:** Innate and adaptive immunity, hypersensitivity, and the stages of osseointegration.
5. **Evaluating Metal Hypersensitivity:** Significance of metal allergies in implants and methodologies for failure prevention.
6. **Dental Materials:** Anatomy of teeth, physical/chemical properties of dental implants, and specific considerations for the oral cavity environment.
7. **Ophthalmic Applications:** Soft and hard contact lenses, intraocular lenses, keratoprosthesis, and high index lenses.
8. **Suture Materials:** Classification (absorbable/non-absorbable, braided/non-braided), tensile strength, and immunogenicity.

9. **Heart and Vascular Implants:** Artificial and tissue heart valves, metal and polymeric stents, and compliance of vascular materials.
10. **Ear Implants:** Middle ear reconstruction, cochlear implants, and mitigation of biofilm formation.
11. **Surface Modifications of Implants:** Physical, chemical, and biological modifications to enhance implant integration and reduce thrombogenicity.